

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 9: REGION DESCRIPTORS

SLO 1: Describe various region-based descriptors used for characterizing object shapes in images.

PICK THE ODD ONE OUT SOLUTIONS

1. Area – Centroid – Perimeter – Orientation

Odd one: **Perimeter**

Reason: Perimeter is a *boundary* descriptor measuring the outer contour length, whereas Area, Centroid, and Orientation are region-based properties.

2. Area – Eccentricity – Compactness – Brightness

Odd one: **Brightness**

Reason: Brightness measures pixel intensity/photometric properties, whereas Area, Eccentricity, and Compactness are geometric region descriptors.

3. Centroid – Orientation – Texture – Area

Odd one: **Texture**

Reason: Texture describes spatial variation of pixel intensities, whereas Centroid, Orientation, and Area are fundamental geometric region descriptors.

4. Region Descriptor – Boundary Descriptor – Shape Descriptor – Color Descriptor

Odd one: **Color Descriptor**

Reason: Color descriptor is a photometric feature space, whereas Region, Boundary, and Shape descriptors specifically characterize geometric shape and spatial structures.

DESCRIPTOR SORTING CHALLENGE SOLUTIONS

Feature	Region Descriptor / Not
Area	Region Descriptor (Quantifies total internal pixels)
Centroid	Region Descriptor (Calculates the center of mass of the region)
Orientation	Region Descriptor (Determines principal axis angle of the region)
Edge gradient	Not a Region Descriptor (Boundary/Gradient feature)
Eccentricity	Region Descriptor (Measures elongation of the region)